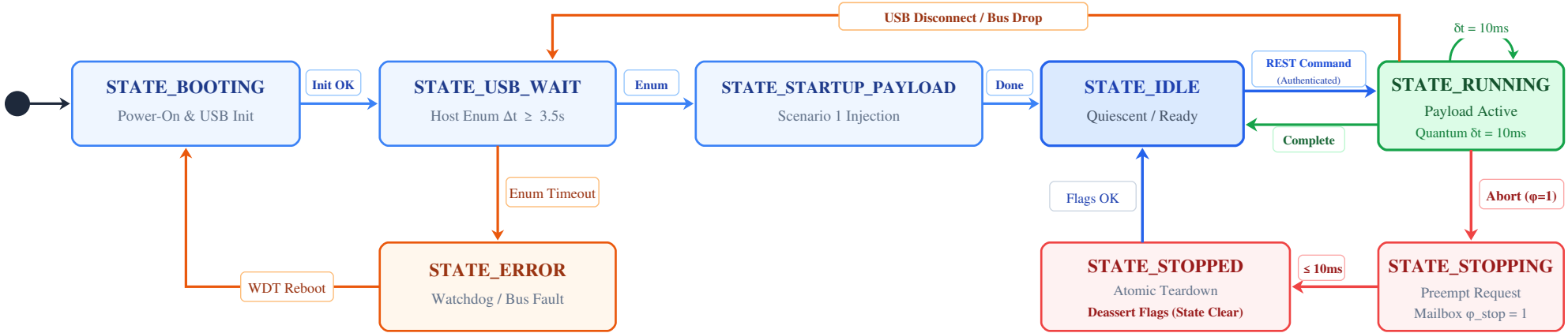

USB Disconnect / Bus Drop
δt = 10ms
STATE_BOOTING
Power-On & USB Init
Init OK
STATE_USB_WAIT
Host Enum Δt ≥ 3.5s
Enum
STATE_STARTUP_PAYLOAD
Scenario 1 Injection
Done
STATE_IDLE
Quiescent / Ready
REST Command
(Authenticated)
STATE_RUNNING
Payload Active
Quantum δt = 10ms
Complete
Enum Timeout
Abort (φ=1)
Flags OK
WDT Reboot
STATE_ERROR
Watchdog / Bus Fault
STATE_STOPPED
Atomic Teardown
Deassert Flags (State Clear)
≤ 10ms
STATE_STOPPING
Preempt Request
Mailbox φ_stop = 1